

Test Coverage Items and Test Cases for Printer Testing

Using Equivalence Partitioning & Boundary Value Analysis (Based on Marketing Info)

1. Input Equivalence Classes

Class ID	Input Parameter	Equivalent Class	Representative / Boundary Value	Test Case ID
I-PHY-1	Physical Interface	USB 2.0 (most common)	USB 2.0 connection	TC-IN-01
I-PHY-2	Physical Interface	Parallel (legacy, but still supported)	Parallel cable	TC-IN-02
I-PHY-3	Physical Interface	Ethernet (network printer)	Wired LAN	TC-IN-03
I-PHY-4	Physical Interface	Wireless (Wi-Fi)	802.11ac	TC-IN-04
I-LOG-1	Logical Interface	PostScript (PS)	PS Level 3 file	TC-IN-05
I-LOG-2	Logical Interface	PCL (HP Printer Command Language)	PCL 6 job	TC-IN-06
I-LOG-3	Logical Interface	ASCII / raw text	Plain text file	TC-IN-07
I-SIZE-1	Print data size	Empty (0 bytes)	0-byte job	TC-IN-08

Class ID	Input Parameter	Equivalent Class	Representative / Boundary Value	Test Case ID
I-SIZE-2	Print data size	Very small (1 byte - 1 KB)	100-byte text	TC-IN-09
I-SIZE-3	Print data size	Typical (1 MB - 10 MB)	5 MB PDF	TC-IN-10
I-SIZE-4	Print data size	Maximum supported (boundary)	Max buffer size (e.g., 256 MB)	TC-IN-11
I-COPIES-1	Number of copies	Minimum (0)	0 copies (should print nothing)	TC-IN-12
I-COPIES-2	Number of copies	Normal (1)	1 copy	TC-IN-13
I-COPIES-3	Number of copies	Many (e.g., 100)	100 copies	TC-IN-14
I-COPIES-4	Number of copies	Maximum (boundary)	Max copies supported (e.g., 999)	TC-IN-15
I-RES-1	Print resolution	Low (150 dpi)	150 dpi	TC-IN-16
I-RES-2	Print resolution	Standard (600 dpi)	600 dpi	TC-IN-17
I-RES-3	Print resolution	High (1200 dpi)	1200 dpi	TC-IN-18

2. Output Equivalence Classes

Class ID	Output Attribute	Equivalent Class	Expected Behavior	Test Case ID
O-QUAL-1	Print quality	Acceptable (no defects)	Sharp, correct colors, no smearing	TC-OUT-01
O-QUAL-2	Print quality	Defective - missing dots	Detectable missing dots on page	TC-OUT-02
O-QUAL-3	Print quality	Defective - banding	Visible horizontal/vertical bands	TC-OUT-03
O-SPEED-1	Print speed	Within specification	First page out time $\leq$ spec	TC-OUT-04
O-SPEED-2	Print speed	Slower than spec	Measure and flag	TC-OUT-05
O-ERROR-1	Error message	Out of paper	Correct error code + user alert	TC-OUT-06
O-ERROR-2	Error message	Paper jam	Correct jam location reported	TC-OUT-07

3. Behavior Equivalence Classes

Class ID	Behavior Scenario	Equivalent Class	Test Case ID
B-NORM-1	Normal printing	Single job, no interruption	TC-BEH-01
B-NORM-2	Normal printing	Multiple sequential jobs	TC-BEH-02

Class ID	Behavior Scenario	Equivalent Class	Test Case ID
B-ERROR-1	Out of paper during job	Printer pauses, shows error, resumes after paper added	TC-BEH-03
B-ERROR-2	Paper jam	Printer stops, user clears jam, job resumes or restarts	TC-BEH-04
B-ERROR-3	Low toner / ink	Warning appears, printing continues (or stops per policy)	TC-BEH-05
B-CANCEL-1	Cancel job	Job removed from queue, no partial print	TC-BEH-06
B-POWER-1	Power cycle during print	Printer recovers, job may restart or cancel gracefully	TC-BEH-07

4. Environment Equivalence Classes

Class ID	Environment Factor	Equivalent Class	Test Case ID
E-OS-1	Operating System	Windows 11 (most common)	TC-ENV-01
E-OS-2	Operating System	macOS Ventura	TC-ENV-02
E-OS-3	Operating System	Linux (Ubuntu)	TC-ENV-03
E-DRV-1	Driver version	Latest WHQL certified	TC-ENV-04
E-DRV-2	Driver version	Legacy driver (previous major release)	TC-ENV-05

Class ID	Environment Factor	Equivalent Class	Test Case ID
E-NET-1	Network mode	Local USB direct	TC-ENV-06
E-NET-2	Network mode	Wired LAN (Ethernet)	TC-ENV-07
E-NET-3	Network mode	Wi-Fi with weak signal	TC-ENV-08
E-MEDIA-1	Paper type	Plain A4	TC-ENV-09
E-MEDIA-2	Paper type	Photo paper (glossy)	TC-ENV-10
E-MEDIA-3	Paper type	Envelope	TC-ENV-11

Detailed Test Cases (English)

TC-IN-01: USB 2.0 Interface

- **Precondition:** Printer connected via USB 2.0, driver installed.
- **Steps:** Send a 1-page Postscript document.
- **Expected:** Prints correctly without errors.

TC-IN-02: Parallel Interface

- **Precondition:** Printer connected via IEEE 1284 parallel cable.
- **Steps:** Send a text file using LPT1 port.
- **Expected:** Document prints fully.

TC-IN-08: Empty print job (0 bytes)

- **Precondition:** Printer idle.
- **Steps:** Submit a print job with 0 data.
- **Expected:** Printer ignores job, no page ejected, no error.

TC-IN-11: Maximum data size (boundary)

- **Precondition:** Printer buffer limit = 256 MB.
- **Steps:** Send a 256 MB PDF file.
- **Expected:** Prints successfully (may spool to disk).

TC-IN-12: 0 copies (boundary minimum)

- **Precondition:** Application allows 0 copies.
- **Steps:** Print 0 copies of a document.
- **Expected:** No output, job completed silently.

TC-IN-15: Maximum copies (boundary)

- **Precondition:** Printer specification says max 999 copies.
- **Steps:** Set copies = 999, print a 1-page document.
- **Expected:** 999 identical pages produced.

TC-OUT-01: Acceptable print quality

- **Precondition:** New toner/ink, default settings.
- **Steps:** Print a test pattern (color gradients, fine lines).
- **Expected:** No missing dots, banding, or color shift.

TC-OUT-06: Out of paper error

- **Precondition:** Tray empty.
- **Steps:** Send a print job.
- **Expected:** Printer displays "Out of paper" on LCD and host PC.

TC-BEH-03: Out of paper during job

- **Precondition:** Paper tray has 1 sheet.
- **Steps:** Print a 3-page document. After page 1 prints, remove remaining paper.
- **Expected:** Printer pauses, shows error. After refill, resumes from page 2.

TC-BEH-07: Power cycle during print

- **Precondition:** Printer printing a 10-page job.
- **Steps:** Unplug power after page 3, wait 10s, plug back.
- **Expected:** Printer restarts, job may be canceled or restarted from page 1 (depending on design). No hardware damage.

TC-ENV-01: Windows 11 environment

- **Precondition:** Windows 11 with latest printer driver.
- **Steps:** Print a Word document via USB.
- **Expected:** Output matches document.

TC-ENV-08: Wi-Fi weak signal (boundary)

- **Precondition:** Printer connected to Wi-Fi, signal strength -75 dBm.
- **Steps:** Send a 10 MB PDF.
- **Expected:** Prints successfully with possible delay, no data corruption.

TC-ENV-11: Envelope media

- **Precondition:** Envelope loaded in manual feed.
- **Steps:** Print address on C5 envelope.
- **Expected:** Correct feed, no crumpling, address aligned.

Marketing-Based Prioritization (Favored Classes)

- **USB 2.0** (I-PHY-1) – 80% of home/office printers use it.
- **Windows 11** (E-OS-1) – most common customer OS.
- **600 dpi** (I-RES-2) – default for most documents.
- **Plain A4** (E-MEDIA-1) – highest volume.
- **PostScript / PCL** (I-LOG-1, I-LOG-2) – required for enterprise compatibility.

These classes should be tested first and most thoroughly.